Probabilistic Modeling and Capacity Planning for GPU-Based AI Services

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1 Denotation

- m: The number GPU;
- N_T: Number of text requests within one minute;
- N_I: Number of image requests within one minute;
- $T_1, T_2, \ldots$: GPU processing time of each text request within one minute;
- $I_1, I_2, \ldots$: GPU processing time of each image request within one minute;
- W: Total GPU workload within one minute.

Accordingly, the total GPU workload can be expressed as:

$$W = \sum_{j=1}^{N_T} T_j + \sum_{k=1}^{N_I} I_k. \tag{1}$$

2 Step 1

2.1 Drawing the Box Plots

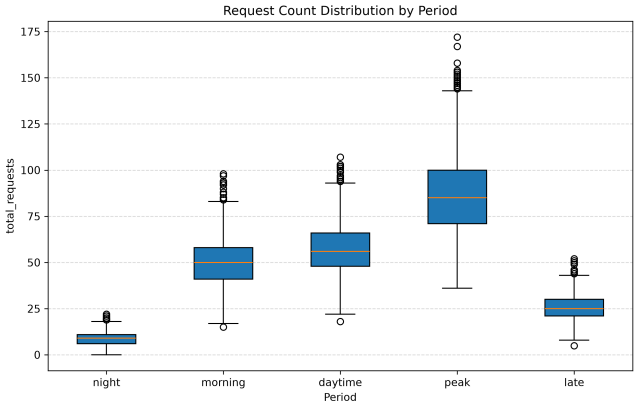


Figure 1: Box plots of text requests

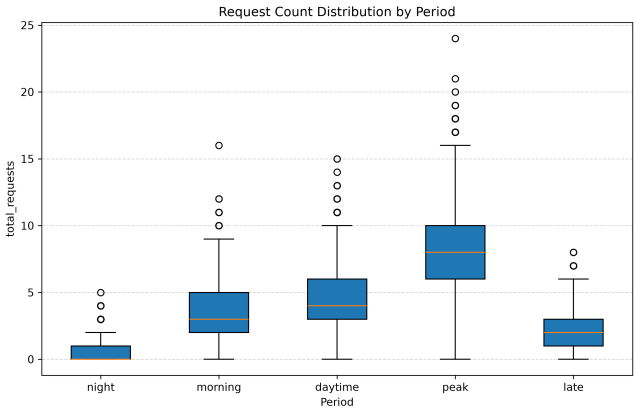


Figure 2: Box plots of image requests

The box plots for the two types of requests are presented in fig. 1 and fig. 2, respectively. As can be observed, an analysis limited exclusively to peak hours on weekdays is insufficient to provide a comprehensive representation of the actual trends in request volumes.

2.2 Establishing Poisson Distributions

We assume that N_t and N_i follow Poisson distributions, which can be expressed as follows:

$$P(N_t = k) = \frac{\lambda_t^k e^{-\lambda_t}}{k!}, P(N_i = k) = \frac{\lambda_i^k e^{-\lambda_i}}{k!}.$$

The parameters λ_t and λ_i can be estimated using the corresponding sample means:

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} N_i.$$

, where n denotes the total minutes.

The 95% confidence interval for the Poisson parameter λ can be approximated using the normal approximation:

$$CI_{95\%} = \left[\hat{\lambda} - 1.96\sqrt{\frac{\hat{\lambda}}{n}}, \hat{\lambda} + 1.96\sqrt{\frac{\hat{\lambda}}{n}} \right].$$

The estimation results are presented in table 1.

Table 1: Parameter Estimation for Poisson Distributions

Type	λ	95% (s)
Text	45.4513	[45.3583, 45.5444]
Image	3.6389	[3.6126, 3.6653]

2.3 Validating the Poisson Distributions

To evaluate the goodness of fit of the Poisson distributions, we conducted a χ^2 goodness-of-fit test, defined as:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i},$$

where O_i denotes the observed frequency of class i, and E_i denotes its corresponding expected frequency. The test results are presented in table 2. The results indicate that the Poisson distributions do not adequately fit the empirical distributions of either text or image requests. This discrepancy may be attributed to the potential dependence among requests within a given minute, which undermines the assumption of independent event occurrences at a fixed time interval. In real-world scenarios, users tend to send multiple messages to AI agents in rapid succession, thereby violating a fundamental assumption of the Poisson model.

A non-homogeneous Poisson process is likely to better fit the empirical request distribution, as its key distinction from the homogeneous Poisson process is that the event arrival rate is allowed to vary over time. Specifically, the number of events occurring within a time interval $[a, b]$ follows a Poisson distribution with a time-dependent parameter:

$$N(b) - N(a) \sim \text{Poisson}\left(\int_a^b \lambda(t)\, dt\right),$$

where $\lambda(t)$ denotes the instantaneous event arrival rate at time t. Unlike the homogeneous Poisson process, which assumes a constant arrival rate, the non-homogeneous Poisson process allows the arrival rate to vary over time, thereby accommodating temporal variations in request intensity.

Table 2: Results of the Poisson Goodness-of-Fit Tests

Type	χ^2	Conclusion
Text	8369296.14	Rejected
Image	72418.63	Rejected

2.4 Correlation Coefficient Between N_t and N_i

The correlation coefficient between N_t and N_i is defined as:

$$r = \frac{\sum_{i=1}^{n}(N_t - \bar{N})(N_i - \bar{N_i})}{\sqrt{\sum_{i=1}^{n}(N_t - \bar{N_t})^2}\sqrt{\sum_{i=1}^{n}(N_i - \bar{N_i})^2}}$$

. The estimated correlation coefficient is approximately 0.7851, indicating a strong positive correlation between text and image request volumes. This suggests that the two types of requests are likely influenced by common traffic patterns, potentially associated with multimodal requests.

3 Step 2

3.1 Report on Service Time Data

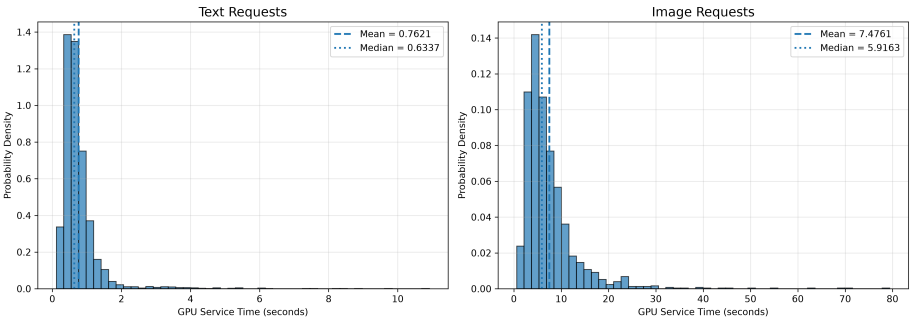


Figure 3: Histograms of the two types of requests

Table 3: Descriptive Statistics of Text and Image Request Durations (seconds)

Type	Size	Mean (s)	Std (s)	Median (s)	95% (s)	99% (s)
Text	4000	0.762147	0.628502	0.633650	1.464310	3.528130
Image	1600	7.476124	6.247750	5.916300	17.529665	29.749451

3.2 Analysis of the Effect of Cold Starts on Service Time

GPU system cold starts can result in substantial increases in response latency. The results presented in table 4 demonstrate that cold starts can considerably extend request processing times. According to the law of total probability:

$$P(T) = P(coldstart)P(t|coldstart) + P(hotstart)P(t|hotstart)$$

, cold starts can contribute to an increase in the overall mean response time and a heavier right tail in the response time distribution.

Table 4: Effect of Cold Starts on Response Time

Type	Cold Start(s)	Hot Start(s)
Text	3.6992	0.6884
Image	23.5168	6.6541

3.3 Establishing the Probability Model

A probability model can be constructed based on the observed service time distributions. In this report, the Gaussian distribution is adopted as the underlying probability model. Specifically:

$$f(T) = \frac{1}{\sqrt{2\pi}\sigma_T} \exp\left(-\frac{(x-\mu_T)^2}{2\sigma_T^2}\right),$$

$$f(I) = \frac{1}{\sqrt{2\pi}\sigma_I} \exp\left(-\frac{(x-\mu_I)^2}{2\sigma_I^2}\right),$$

where T and I denote the service durations of individual text and image requests, respectively. Here, μ_T and μ_I represent the mean service times, while σ_T^2 and σ_I^2 denote the corresponding variances. Furthermore, $E(T^2)$ and $E(I^2)$ represent the second raw moments of the respective service time distributions, which are related to the means and variances by:

$$E(T^2) = \sigma_T^2 + \mu_T^2, \qquad E(I^2) = \sigma_I^2 + \mu_I^2.$$

3.4 Estimating the Model Parameters

The estimated model parameters are presented in table 5.

Table 5: Estimated Parameters of the Gaussian Distributions

μ_T	σ_T^2	μ_I	σ_I^2	$E(T^2)$	$E(I^2)$
0.7621	0.6285	7.4761	6.2477	0.9758	94.9024

4 Step 3

4.1 Estimation of $E(W)$ and $D(W)$

Assuming N_T and N_I are independent to each other, We have:

$$\begin{aligned} E(W) &= E\left(\sum_{j=1}^{N_T} T_j\right) + E\left(\sum_{k=1}^{N_I} I_k\right) \\ &= E(N_T)E(T) + E(N_I)E(I) \\ &= \lambda_T \mu_T + \lambda_I \mu_I. \end{aligned}$$

and

$$\begin{aligned} D(W) &= D\left(\sum_{j=1}^{N_T} T_j\right) + D\left(\sum_{k=1}^{N_I} I_k\right) \\ &= E(N_T)D(T) + E(N_I)D(I) \\ &= E(N_T)E(T^2) + E(N_I)E(I^2). \end{aligned}$$

The estimated results are presented on table 6.

Table 6: Estimated Parameters of the Gaussian Distributions

Target	Estimation	Contribution of T	Contribution of I
$E(W)$	61.845	34.63	27.22
$D(W)$	389.6947	44.34	345.44

4.2 Assumption of Dependence Between N_T and N_I

The assumption of independence between N_T and N_I may not hold in practice. As indicated by the estimated correlation coefficient, the two types of requests exhibit a positive correlation, implying that $Cov(N_T, N_I) \neq 0$. Ignoring this dependence may introduce errors into the estimation of the total workload variance. Specifically, the variance calculation should account for the covariance between the two request types.

In the following report we will follow the assumption of independence.

5 Step 4

5.1 Estimation of W Using The Central Limit Theorem

According to the central limit theorem,

$$\frac{W - E(W)}{\sqrt{D(W)}} = \frac{\sum_{j=1}^{N_T} T_j + \sum_{k=1}^{N_I} I_k - E(W)}{\sqrt{D(W)}} \sim \mathcal{N}(0,1)$$

.

5.2 Estimation of m

To estimate m, which is subject to the condition $P(W > 48m)$, we can use the following approximation:

$$P(W > 48m) = P(\hat{W} > \frac{48m - E(W)}{\sqrt{D(W)}}) = 0.01$$

, where

$$\hat{W} = \frac{W - E(W)}{\sqrt{D(W)}}$$

, namely

$$\frac{48m - E(W)}{\sqrt{D(W)}} = 2.33$$

and $\hat{W} \sim \mathcal{N}(0,1)$. The resulting estimate is $m = 2.2467$.

5.3 Rationale and Risks of Using the Central Limit Theorem

When both N_T and N_I are sufficiently large, W can be approximately modeled by a normal (Gaussian) distribution according to the Central Limit Theorem. However, potential risks arise from extreme values in the tails of the data distribution, which may lead to significant deviations from the normal approximation.

Based on the estimated values of $N_T = 45.45$ and $N_I = 3.64$, the sample set is considered sufficient. Nevertheless, the imbalance between text and image requests may introduce instability into the normal approximation for image requests. Furthermore, the long-tailed distribution of service times caused by cold starts may affect the accuracy of the approximation, particularly in the right tail.

6 Step 5

6.1 Monte Carlo Simulation Test

To validate the accuracy of the Gaussian distribution model, a Monte Carlo simulation is conducted using randomly generated requests. The simulation consists of 500,000 iterations, with each iteration representing a one-minute interval. During each iteration, the total request processing time is calculated based on the estimated values of N_T and N_I. An iteration is recorded as a failure if the total request processing time within that one-minute interval exceeds the GPU capacity of $48m$.

6.2 Test result

The results $ratio = 0.9726\%$ indicate that the failure ratio remains below 1%, thereby satisfying the specified requirement.

7 Step 6

7.1 Scene A

Assuming that the number of text and image requests rise by 30%, while the time of single text or image request remains unchanged, the number of GPU m should be updated. Specifically,

$$N_T = N_T \cdot 130\%$$

$$N_I = N_I \cdot 130\%$$

. If m remains unchanged, the Monte Carlo Simulation Test suggests that the ratio of GPU overload will rise to 11.12%. The re-evaluation of m based on the new situation is approximately 2.7676.

7.2 Scene B

Assuming that the likelihood of image request is changed to 35%, specifically,

$$N_I = (N_T + N_I) \cdot 35\%$$

$$N_T = (N_T + N_I) \cdot 65\%$$

. If m remains unchanged, the Monte Carlo Simulation Test suggests that the ratio of GPU overload will rise to 86.46%. To maintain the 1% requirement, m should be adjusted to 5.1615.

8 Step 7

8.1 Decision Memorandum

- Minimum configuration required to meet the current service-level targets: $m = 2.2467$;

- Actual recommended configuration: $m = 3$;
- Recommended configuration after three months: $m = 3$;
- When the proportion of image requests increases to 35%, simply adding GPU machines is expected to become impractical, since $I = 9.81T$. The estimated value of m will increase sharply as I increases.
- The risk of GPU overload arises primarily from image request processing time and cold-start latency, whereas request fluctuations and average data traffic have comparatively minor contributions.
- It is recommended that the GPU service platform continuously monitor the average number of image requests and update the estimation of W in real time. A GPU capacity expansion warning should be triggered if the following indicator falls below 2.33:

  $$\frac{48m - E(W)}{\sqrt{D(W)}} < 2.33,$$

  as this would indicate a degradation in the 1% requirement.
- The conclusions of this report rely on the independence assumption between N_T and N_I, as well as the Gaussian distribution model established based on prior knowledge. The independence assumption between N_T and N_I warrants further investigation through continuous data collection, particularly given the increasing popularity of multimodal LLMs.